MINING DEALER BEST PRACTICE

Bucket Stop Measuring Device for 994H Wheel Loader

Maintenance and Repair | Application | Component Life Management | Component Renewal (CRC) | MARC Management

February 2023
1.01-230221-1

1.0 Introduction

When measuring the rear bucket stops that protect the tilt cylinders when the bucket is in the racked back position, the attachments need to be moved, causing difficulty and inaccuracy in the measurement values performed by the technician.

This best practice provides a solution that significantly optimizes the technician's bucket stop measurement activity.

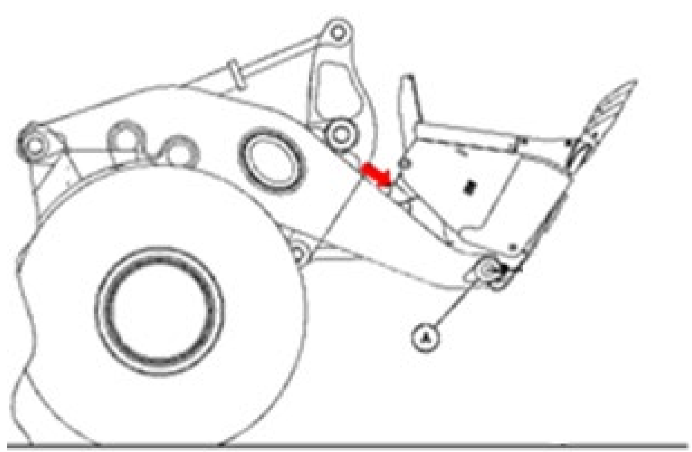
2.0 Best Practice Description

One of the main machines responsible for production in a mining industry is the wheel loader. These versatile machines move and articulate the implements with ease. During loading where the implement is used, one of the possible movements is the racked back position. In this position the bucket is fully retracted, allowing the tilt cylinders to be fully opened to the point of reaching their end of stroke.

This end-of-stroke condition of the tilt cylinders can damage the cylinders and rods if pressure is applied to the cylinder or bucket when they are in this position. Caterpillar's recommendation is to avoid this condition and to apply stops at the rear of the bucket, so the stops will touch the lift arm before the tilt cylinders reach their end of stroke during the racked back position.

The rack back stops should be checked prior to installing a different bucket, lift arms or tilt cylinders. The standardization of the stops is done by measurements between the bucket and the lifting arm, recommended in the procedure "Bucket Stops Provide Additional Protection for Tilt Cylinder (i08683630)".

During this activity, improvement points were identified to optimize the measurement and assertiveness.



Picture 1 – Bucket stop location.

The device consists of a rod (Picture 2), a magnetic base, and a laser tape measure:

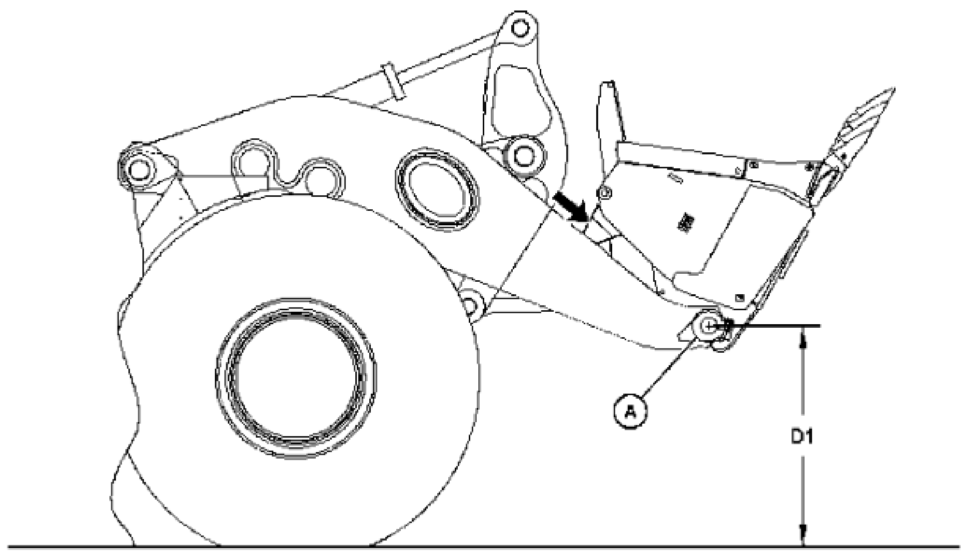
Picture 2 – Installed device.

3.0 Implementation Steps

1. Following the recommendations for measuring stops contained in the procedure "Bucket Stops Provide Additional Protection for Tilt Cylinder (i08683630)" and shown in Table 1, there is a necessity to keep the center of the bucket pin at a height of 2.524mm from the ground (D1) (Picture 3), with the bucket in the racked back position:

Machine	(D1) (Bucket Pin Height at Off Rack)
994H EHL	2524 mm (8.28 ft)

Table 1 – Reference table.



Picture 3 – Distance of the ground.

2. To perform the measurement using the device, the rod must be installed in the bucket pin with the aid of a bolt in the same specifications as the internal thread of the bucket pin according to Picture 4:

Picture 4 – Rod installed in the bucket pin.

3. In the center of the head of this screw, a boss should be adapted for mounting the rod. A movable bearing should be installed at the end of the rod. This mobile bearing will be responsible for fixing the rod in the bolt applied to the bucket pin, so that the rod is rotational.

4. The rod should have a length of 2.524mm (Picture 5) and has the function of referencing the measure D1, according to the recommendations contained in the literature i08683630. Therefore, the rod will replace the need for measurement between the bucket and the ground.

Picture 5 – Length of the rod.

5. Magnetics bases are used to hold the laser tape measure during measurement and should be installed at the rear of the bucket (Picture 6).

Picture 6 – Magnetics bases installed.

6. The laser measuring tape takes the measurements, and the values are sent via Bluetooth to a cell phone application.

4.0 Benefits

- Better ergonomics when taking measurements.
- Greater assertiveness in taking measurements.

5.0 Resources Required (per machine)

Labor for installing the tool on the machine: 10 minutes.
Necessary parts:

- Rod carbon steel SAE1045 (2.524mm) x 10mm.
- Laser tape measurement tool.
- Magnetic base.

6.0 Supporting Attachments / References

"Bucket Stops Provide Additional Protection for Tilt Cylinder - M0064430-05 - i08683630".

7.0 Related Best Practices

N/A

8.0 Acknowledgements

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